

MICROSOFT GARAGE

WORKSHOP

2026 INTERN COHORT

From ~~Vibe Coding~~ to *Agentic Engineering*.

The leap from prompting a toy to co-architecting a real codebase.

NOIDA • HYD • BLR • _

A LITTLE ABOUT ME

A little *about me.*

◇ Microsoft IDC (2021 – Present)

Applied Scientist II (2025 –) · M365 Copilot Extensibility

Software Engineer II (2024 – 25) · M365 Copilot Extensibility

Data & Applied Scientist II (2023 – 24) · Bing Autosuggest + Search History

Data & Applied Scientist (2021 – 23) · Bing Autosuggest

◇ Visiting Faculty · Bower School of Entrepreneurship

◇ IIT Bombay (2017 – 21)

Institute Rank 2

Major · Mechanical Engineering · 9.90/10

Minor · Computer Science & Engineering

◇ Author · **Beyond Code** (National Bestseller)

◇ Author · **Low-Pass Filter** — an AI newsletter

◇ 7 US Patents · 2 Research Papers

◇ Empowered **20,000+** students & professionals in AI & Data Science



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tw [@tezansahu](https://twitter.com/tezansahu)

web tezansahu.com

THE CREW MAKING THIS HAPPEN

Meet the *volunteers*.

Stuck for two minutes in Part 2? Wave one of them over.



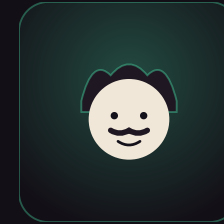
Syed

#01 • VOLUNTEER



Karan

#02 • VOLUNTEER



Mohit

#03 • VOLUNTEER



VijayLaxmi

#04 • VOLUNTEER



Tejj

#05 • VOLUNTEER



Gaurav

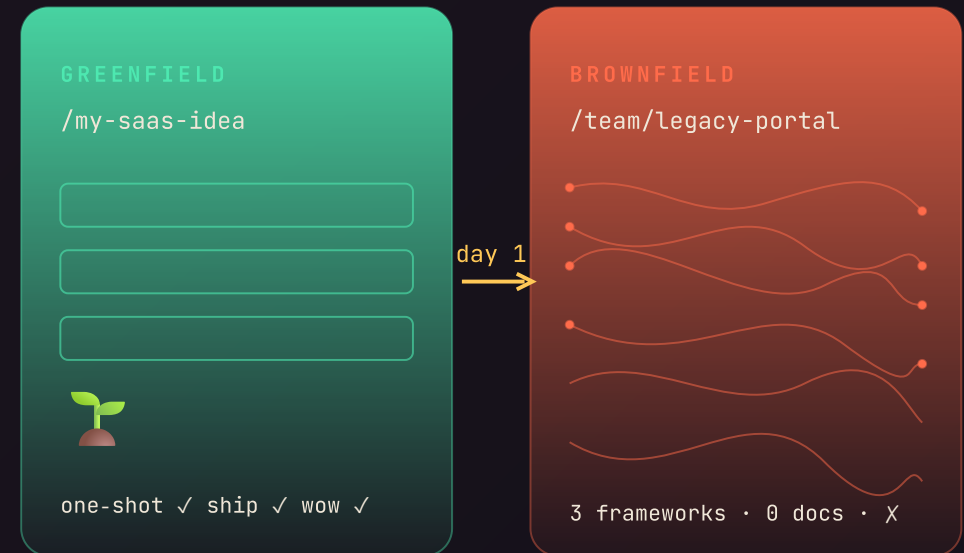
#06 • VOLUNTEER

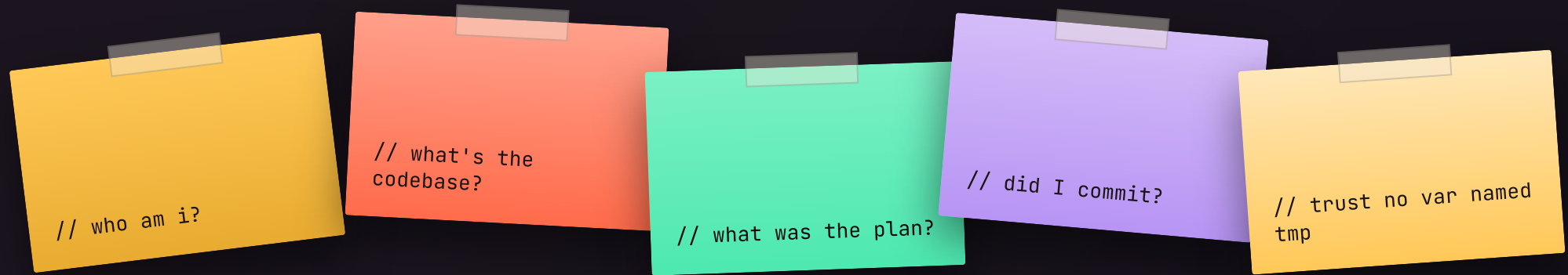
THE WALL EVERYONE HITS

Vibe coding has a ceiling.

Greenfield demos lie to you.

Real teams ship into brownfield.





AI is a brilliant new starter with no memory.

Every agent run is *Memento*. Your job is the sticky notes.

BEFORE

```
you = engineer  
ai = autocomplete
```

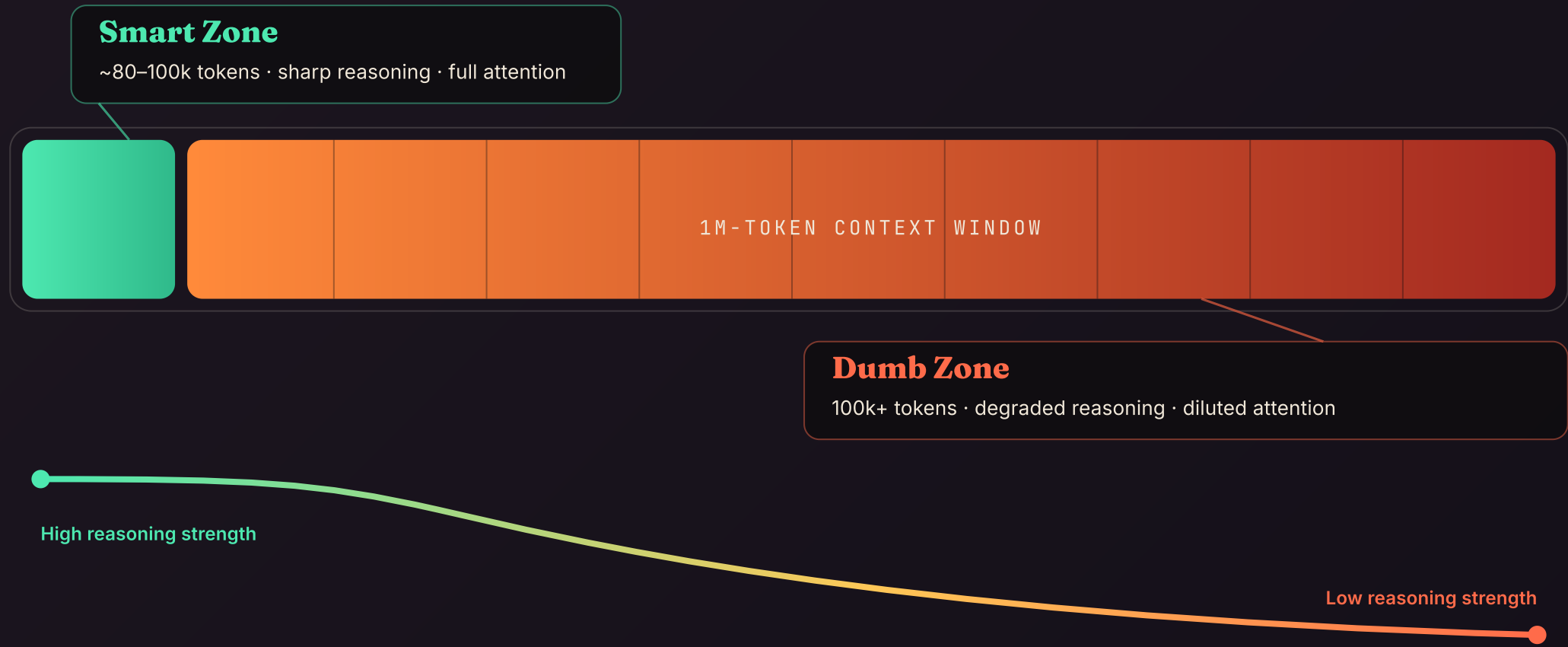


AFTER

```
you = architect + PM  
ai = staff engineer
```

You don't type code.
You **commission** it.

1M tokens of context. ~100k tokens of thinking.



why?

Attention is **quadratic**. Every token relates to every other — n^2 scaling. More tokens dilute focus.

trap

Bigger context windows ship **more dumb zone**. Great for retrieval. Not for reasoning.

fix

Keep tasks **inside the smart zone**. Externalize state to files. `/clear` early, often.

BEFORE WE INSTALL OURS · WHAT'S ALREADY IN THE BOX

GitHub Copilot's **toolbox** / a quick tour.

You probably opened chat and started typing. There are **eight** levers most people never touch.

`model picker`

Pick the brain.

GPT-5.2 · Claude Sonnet · Gemini ·
per-task swap.

`agent mode`

Autonomous loop.

Plan → tool → edit → verify → repeat.

`plan mode`

Read-only think.

Drafts the plan. No writes. Review
first.

`agents.md`

Always-on context.

`.github/` · system prompt for the
repo.

`*.instructions.md`

Scoped rules.

`applyTo: '**/*.tsx'` · per-path
conventions.

`*.prompt.md`

Saved slash cmds.

`/review-pr` · reusable prompts with
args.

`*.agent.md`

Custom personas.

Scoped tools, own model, agent-to-
agent handoffs.

`mcp servers`

Plug in tools.

Jira · DBs · Playwright · your APIs.

\$ All eight are **knobs**. Each tuned in isolation. None of them answer: *how do I make the agent run this workflow reliably, every time?* → **skills**.

REUSABLE INSTRUCTIONS FOR THE AMNESIAC ENGINEER

Agent Skills / the onboarding doc.

/grill-me

Interview me until we share understanding.

/write-a-prd

Crystallize the convo into a PRD.

/prd-to-issues

Slice the PRD into tracer-bullet issues.

/tdd

Red → green → refactor, on a loop.

`.claude/skills/`

OR

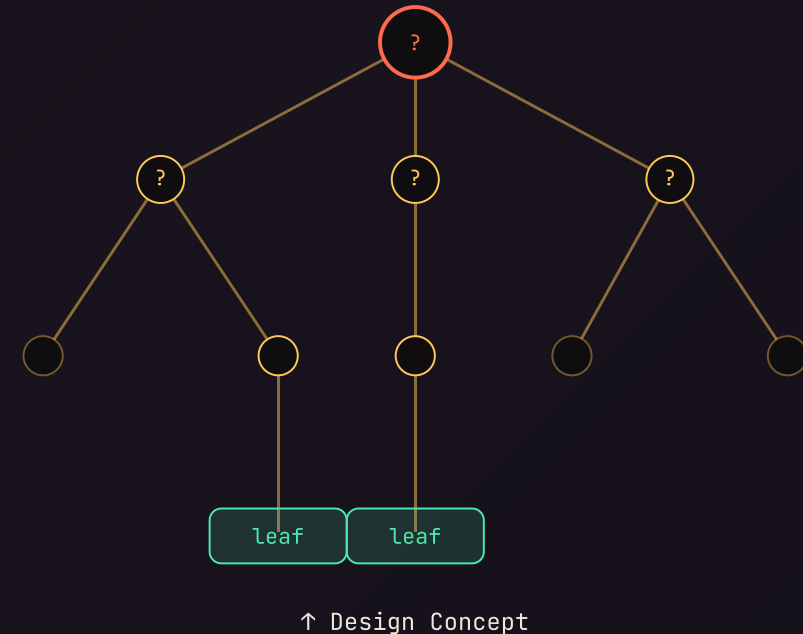
`.agents/skills/`

```
└─ <skill-name>/  
  └─ SKILL.md — what · when · how  
  └─ references/ — docs the agent can pull  
  └─ scripts/ — executable helpers
```

Every answer prunes a branch.

The leaf is your **Design Concept** — the shared map between you and the AI.

- ? What is a "correlation" here?
- ? How many data points to trust it?
- ? Strong vs weak — how do we show it?
- ? Dismissible? Sensitive? Causal?



Plan mode vs. Grilling.

PLAN MODE

AI → you

*“Here's my plan.
Approve?”*

ai projects its assumptions onto your problem.

VS

GRILLING

you ← AI

*“What counts as a
correlation?”*

ai extracts your intent. you stay the source of
taste.

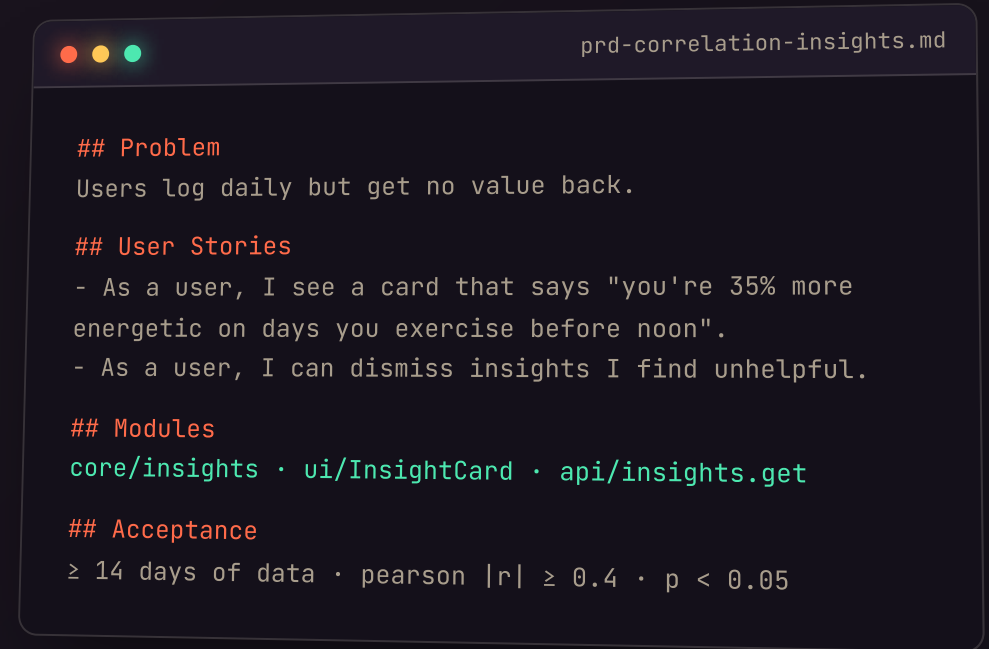
You are the domain expert. **Make the AI ask.**

CRYSTALLIZE THE CONVERSATION

The PRD is your externalized brain.

Convo → document. Now any fresh agent can pick up where you left off.

```
if (session.crashes) { newAgent.read("PRD.md") }
```



```
## Problem
Users log daily but get no value back.

## User Stories
- As a user, I see a card that says "you're 35% more energetic on days you exercise before noon".
- As a user, I can dismiss insights I find unhelpful.

## Modules
core/insights · ui/InsightCard · api/insights.get

## Acceptance
≥ 14 days of data · pearson |r| ≥ 0.4 · p < 0.05
```

PRAGMATIC PROGRAMMER · 1999

Don't build a layer. Fire one bullet.

A thin vertical slice through every layer, end-to-end.
Then expand.

“Stop letting the agent outrun its headlights.”

HORIZONTAL · slop

UI · all 6 cards

API · all endpoints

Logic · 12 correlations

Storage · new schema

VERTICAL · tracer

UI · 1 card

API · 1 endpoint

Logic · 1 correlation

Storage · 1 table

↳ nothing works · 1000 LOC to review ↳ end-to-end · validated · ship

A KANBAN FOR YOUR FLEET OF AMNESIACS

PRD → queue of vertical slices.

 TODO

#001 • TRACER

1 correlation, end-to-end

ui • api • logic • storage

#002

Add 4 more correlations

#003

Strength badges (weak / strong)

#004

Dismiss insight + persist

 DOING

#001

tracer bullet • TDD red

ralph • agent-0

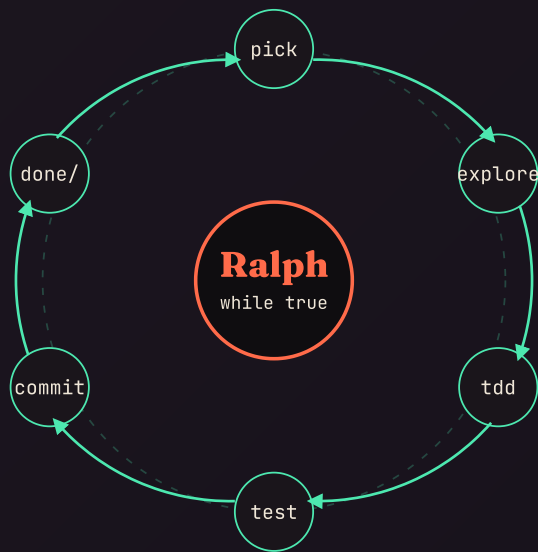
 DONE

#000

install skills • seed data

Meet Ralph.

A bash loop. A fresh agent. One issue at a time. Until NO MORE TASKS.



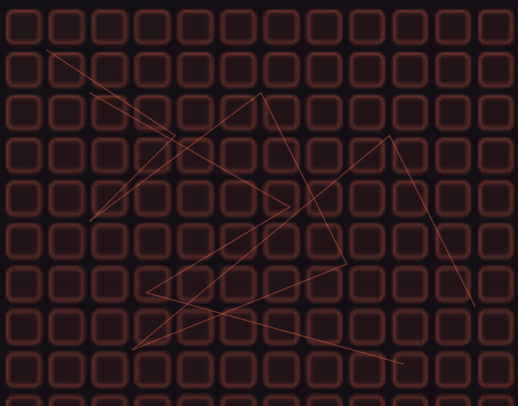
```
$ ./afk.sh 20
> iter 1/20 · picking #001 · tracer bullet
> green · committed · moved to done/
> iter 2/20 · picking #002
> green · committed · moved to done/
> iter 8/20 · NO MORE TASKS · exit 0
```

Make your modules **deep**, not **shallow**.

ousterhout · a philosophy of software design —

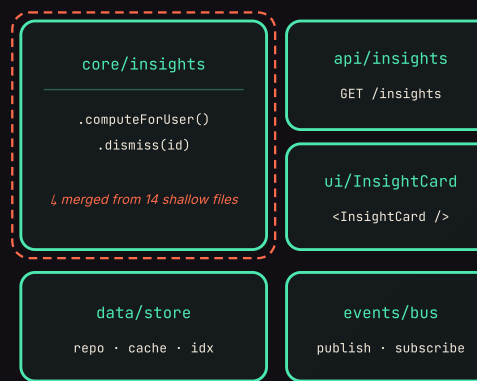
Small interface, rich behavior. The shape your agent can navigate.

shallow
many tiny files



- big API · thin behavior
- logic spread across the maze
- agent traces graphs, eats context

deep
few modules · thick logic



- small API · thick implementation
- boundary = test seam
- agent reasons by interface

brownfield play → find a coupled shallow cluster · wrap a test around the boundary · merge internals · keep tests green.

There's a skill for that: [/improve-codebase-architecture](#)

SCALING WITHOUT BURNING OUT

Grey-box modules.



You design the boundary. AI fills the inside. Tests keep it honest.

Heavy on **design**. Light on **execution**.

01

brief

you

02

grill

you + AI

03

prd

AI · you review

04

issues

AI · **you review**

05

ralph

AI · afk

06

ship

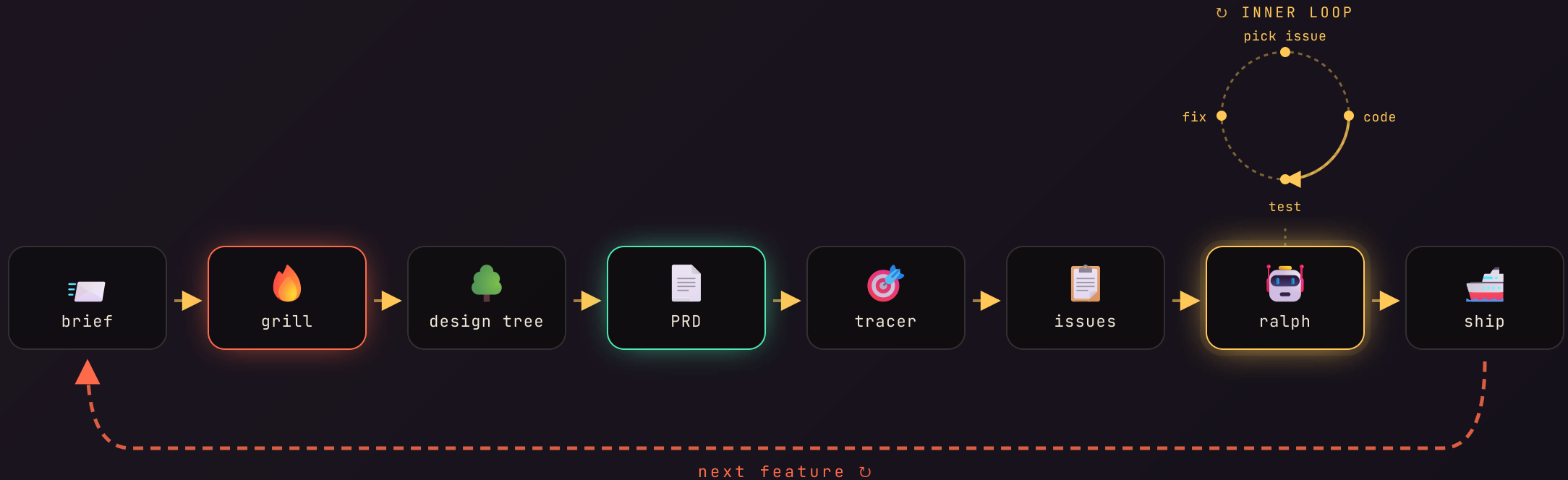
you skim

Review the **issues**, not the diffs.

SCREENSHOT THIS

The agentic engineering loop.

You spin the outer loop. Ralph spins the inner one.



● YOU STEER · TASTE ● ARTIFACTS · CONTRACTS ● RALPH · AFK · SPINNING

Your turn.

01

`git clone`

the pulse repo

02

`cat briefs/0?-*.md`

pick one

03

`/grill-me`

until you can defend a design

04

`/write-a-prd → /prd-to-issues`

queue it up

05

`./ralph/once.sh`

let it rip • stretch goal

HOUSE RULE

No code until you can show your design concept. **No exceptions.**

Stop vibing. Start **engineering.**

Software fundamentals matter more than ever.

READ

5 Agent Skills I Use Every Day

Tracer Bullets · keeping slop under control

Codebases AI Agents Love

9 Ways AI Coding Rewired My Brain

The /grill-me skill

CLASSICS

Brooks · *The Design of Design*

Hunt & Thomas · *The Pragmatic Programmer*

Ousterhout · *A Philosophy of Software Design*



connect on **LinkedIn**

linkedin.com/in/tezan-sahu



read **Low-Pass Filter**

lowpassfilter.substack.com